

Lab 3 / Project 2

Autonomous Vehicle: Putting it All Together

COMPENG 4DS4 - Winter 2026

McMaster University

Group 19

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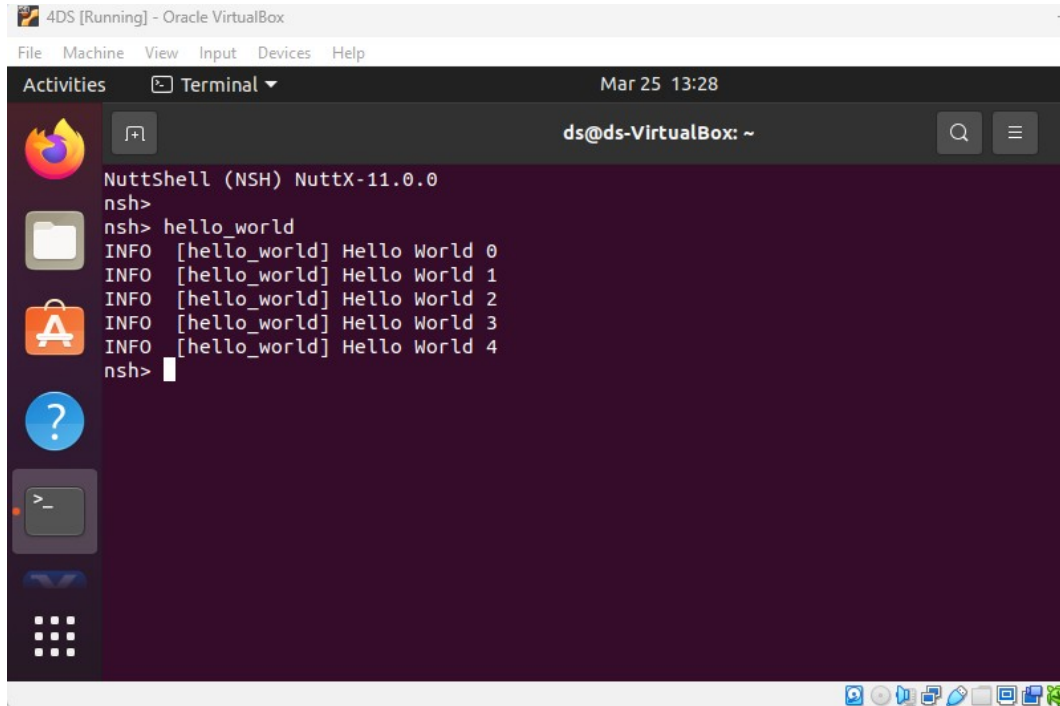
Declaration of Contributions

Ali Raza: Experiment 1 (Build and program PX4). Experiment 2 (hello_world application on NuttX). Experiment 3A (uORB messaging, subscribe and publish).

Zuhaib Quraishi: Experiment 3C (motor control via test_motor). Experiment 4 (MAVLink setup and communication). Project 2 Part 1 (RC controller motor and servo control).

Khushi Akbari: Experiment 5 (ultrasonic sensor on Raspberry Pi). Experiment 6 (camera direction detection). Project 2 Part 2 (autonomous control via MAVLink).

Experiment 2B: Hello World Application



4DS [Running] - Oracle VM VirtualBox

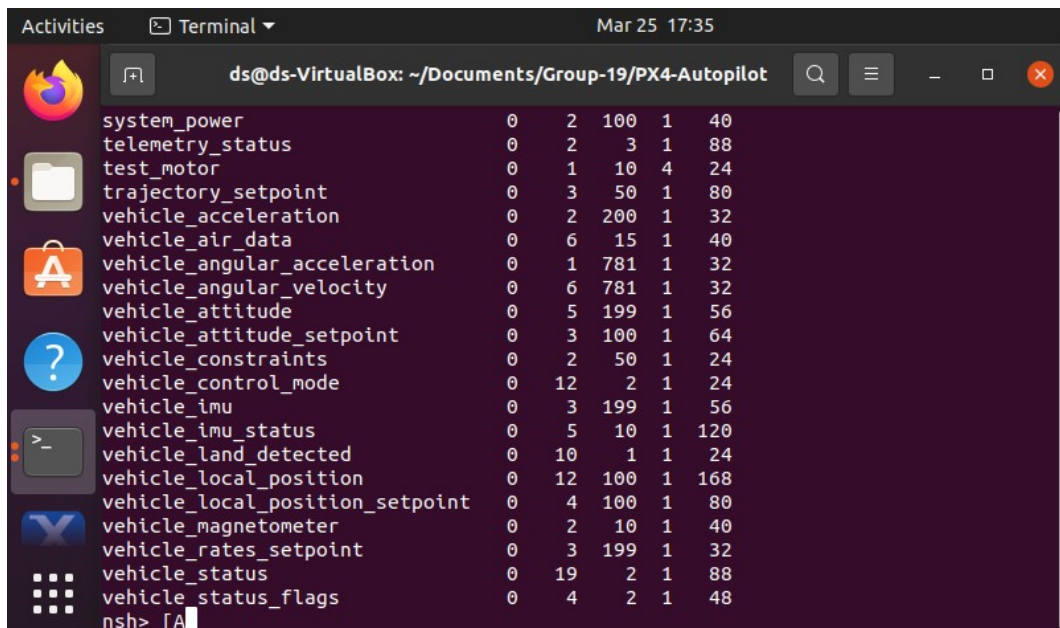
File Machine View Input Devices Help

Activities Terminal Mar 25 13:28

ds@ds-VirtualBox: ~

```
NuttShell (NSH) NuttX-11.0.0
nsh>
nsh> hello_world
INFO [hello_world] Hello World 0
INFO [hello_world] Hello World 1
INFO [hello_world] Hello World 2
INFO [hello_world] Hello World 3
INFO [hello_world] Hello World 4
nsh>
```

Experiment 3A: uORB Top



Activities Terminal Mar 25 17:35

ds@ds-VirtualBox: ~/Documents/Group-19/PX4-Autopilot

```
system_power          0    2  100  1  40
telemetry_status      0    2    3  1  88
test_motor            0    1   10  4  24
trajectory_setpoint   0    3   50  1  80
vehicle_acceleration  0    2  200  1  32
vehicle_air_data      0    6   15  1  40
vehicle_angular_acceleration 0    1  781  1  32
vehicle_angular_velocity 0    6  781  1  32
vehicle_attitude      0    5  199  1  56
vehicle_attitude_setpoint 0    3  100  1  64
vehicle_constraints    0    2   50  1  24
vehicle_control_mode   0   12    2  1  24
vehicle_imu            0    3  199  1  56
vehicle_imu_status     0    5   10  1 120
vehicle_land_detected  0   10    1  1  24
vehicle_local_position 0   12  100  1 168
vehicle_local_position_setpoint 0    4  100  1  80
vehicle_magnetometer   0    2   10  1  40
vehicle_rates_setpoint 0    3  199  1  32
vehicle_status         0   19    2  1  88
vehicle_status_flags   0    4    2  1  48
nsh> [A]
```

Experiment 3A: Listener sensor_combined

```
Activities Terminal Mar 25 17:36
ds@ds-VirtualBox: ~/Documents/Group-19/PX4-Autopilot

vehicle_land_detected      0  10  1  1  24
vehicle_local_position     0  12 100  1 168
vehicle_local_position_setpoint 0  4 100  1  80
vehicle_magnetometer       0  2  10  1  40
vehicle_rates_setpoint     0  3 199  1  32
vehicle_status             0 19  2  1  88
vehicle_status_flags       0  4  2  1  48
nsh> q
nsh: q: command not found
nsh> listener sensor_combined

TOPIC: sensor_combined
sensor_combined
timestamp: 233544303 (0.009789 seconds ago)
gyro_rad: [0.0014, 0.0052, 0.0078]
gyro_integral_dt: 5000
accelerometer_timestamp_relative: 0
accelerometer_m_s2: [2.4932, 0.1514, -10.5951]
accelerometer_integral_dt: 4997
accelerometer_clipping: 0
nsh>
```

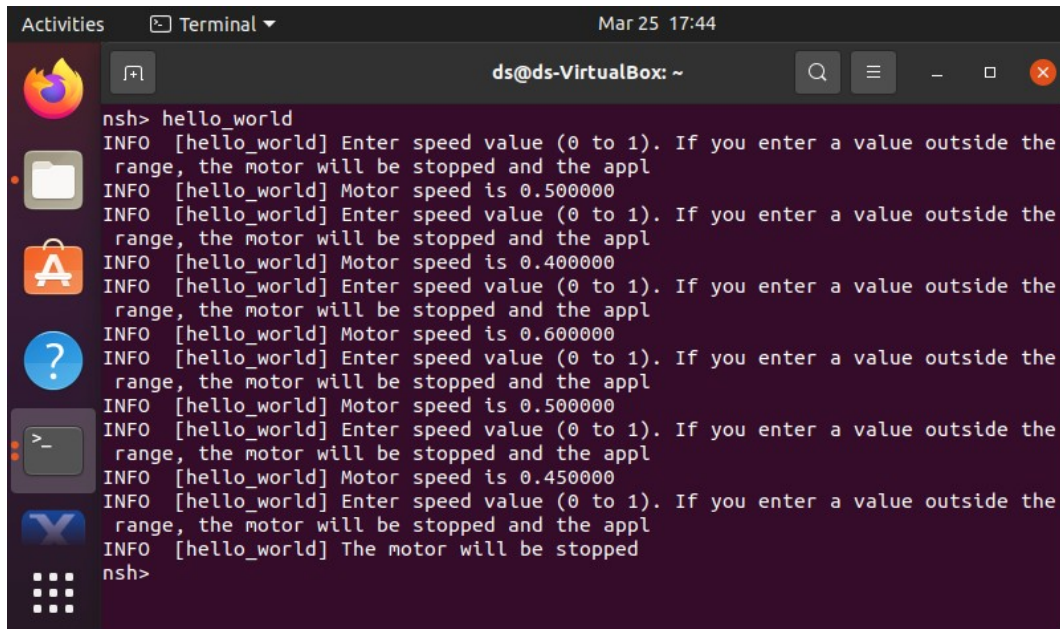
Experiment 3A: Listener input_rc

```
Activities Terminal Mar 25 17:37
ds@ds-VirtualBox: ~/Documents/Group-19/PX4-Autopilot

vehicle_magnetometer       0  2  10  1  40
vehicle_rates_setpoint     0  3 198  1  32
vehicle_status             0 19  2  1  88
vehicle_status_flags       0  4  2  1  48
nsh> listener input_rc

TOPIC: input_rc
input_rc
timestamp: 337084748 (0.001018 seconds ago)
timestamp_last_signal: 337084748
rssi: 255
rc_lost_frame_count: 0
rc_total_frame_count: 0
rc_ppm_frame_length: 0
values: [1514, 1514, 1514, 1514, 1024, 1514, 1514, 1514, 1514, 1514,
1514, 1514, 1514, 1514, 1514, 998, 998]
channel_count: 18
rc_failsafe: False
rc_lost: False
input_source: 9
nsh>
```

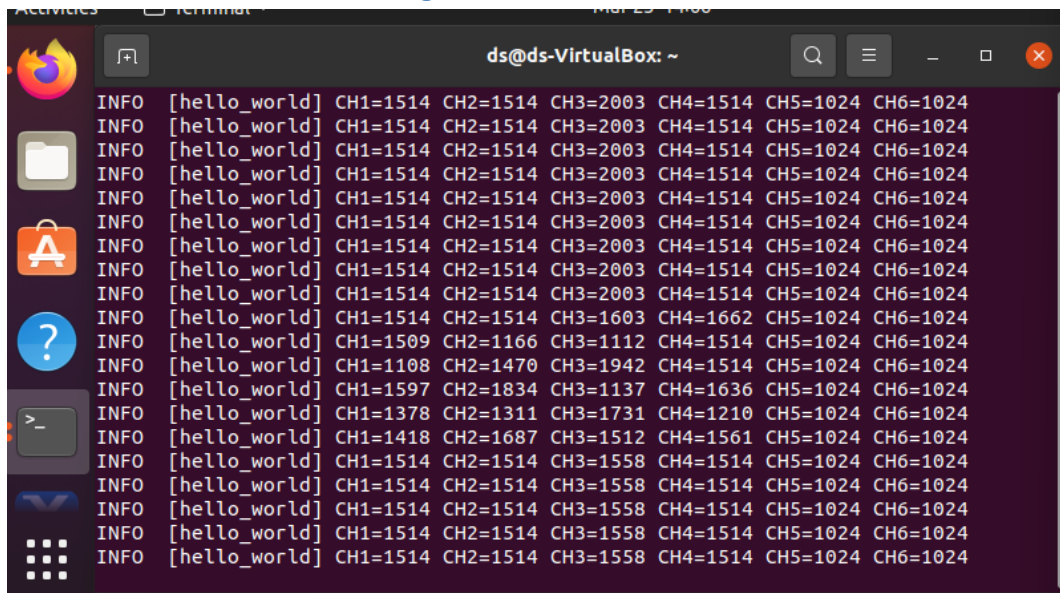
Experiment 3C: Motor Speed Control



A terminal window titled 'ds@ds-VirtualBox: ~' showing the execution of a program named 'hello_world'. The program prompts the user to enter a speed value between 0 and 1. The user enters several values, and the program outputs the corresponding motor speed. The output shows that the motor speed is 0.500000 for a speed value of 0.5, 0.400000 for 0.4, and 0.600000 for 0.6. The program also includes a warning message: 'INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the range, the motor will be stopped and the appl'. The terminal window has a dark background and a sidebar with application icons.

```
nsh> hello_world
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] Motor speed is 0.500000
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] Motor speed is 0.400000
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] Motor speed is 0.600000
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] Motor speed is 0.500000
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] Motor speed is 0.450000
INFO [hello_world] Enter speed value (0 to 1). If you enter a value outside the
range, the motor will be stopped and the appl
INFO [hello_world] The motor will be stopped
nsh>
```

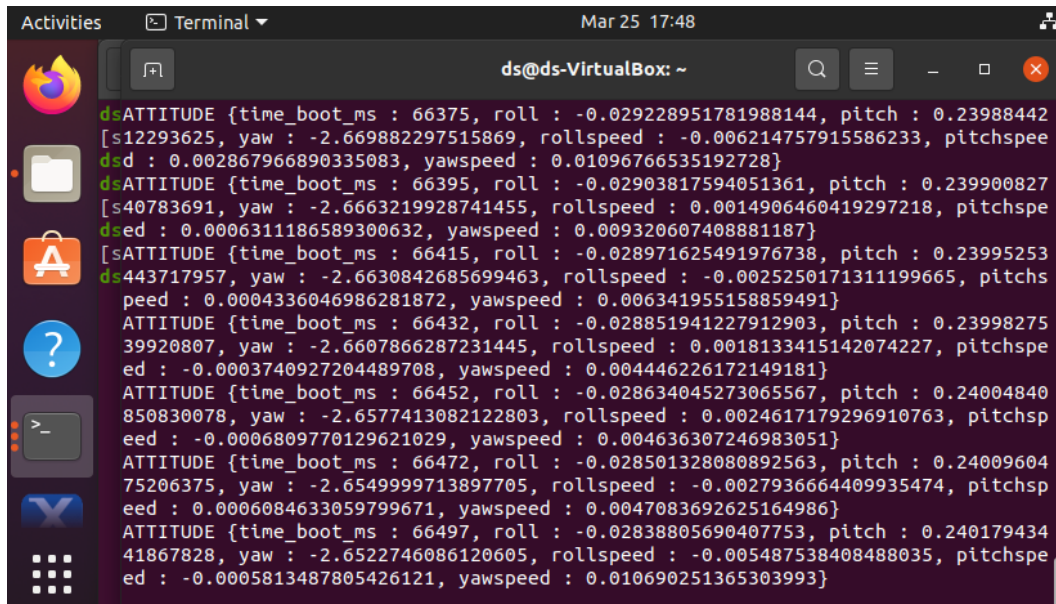
Experiment 4A: MAVLink Message Stream



A terminal window titled 'ds@ds-VirtualBox: ~' showing the execution of a program named 'hello_world'. The program outputs a stream of MAVLink messages. Each message is a line of text starting with 'INFO [hello_world]' followed by six channels (CH1 to CH6) and their values. The values are integers. The terminal window has a dark background and a sidebar with application icons.

```
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=2003 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1603 CH4=1662 CH5=1024 CH6=1024
INFO [hello_world] CH1=1509 CH2=1166 CH3=1112 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1108 CH2=1470 CH3=1942 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1597 CH2=1834 CH3=1137 CH4=1636 CH5=1024 CH6=1024
INFO [hello_world] CH1=1378 CH2=1311 CH3=1731 CH4=1210 CH5=1024 CH6=1024
INFO [hello_world] CH1=1418 CH2=1687 CH3=1512 CH4=1561 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1558 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1558 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1558 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1558 CH4=1514 CH5=1024 CH6=1024
INFO [hello_world] CH1=1514 CH2=1514 CH3=1558 CH4=1514 CH5=1024 CH6=1024
```

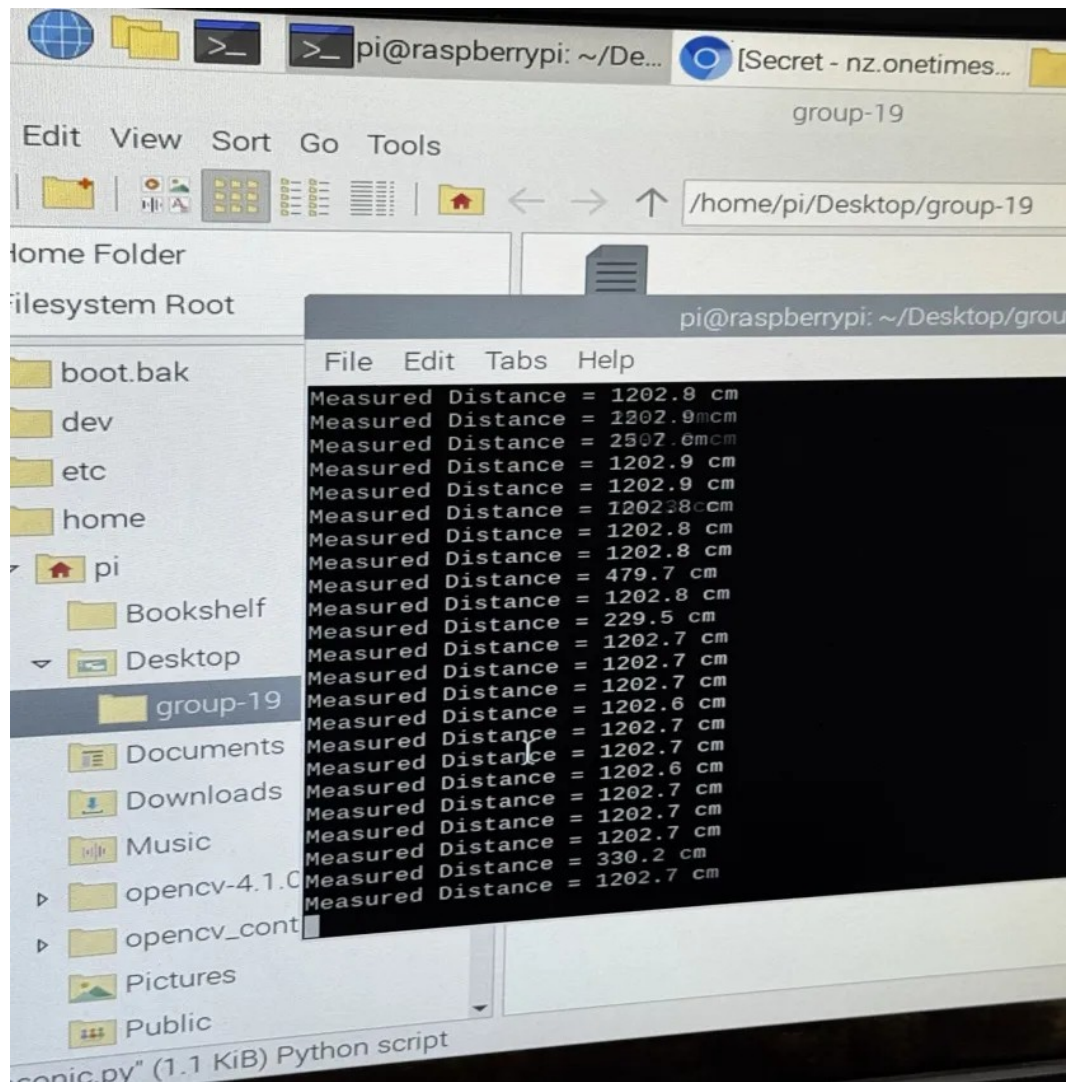
Experiment 4A: ATTITUDE Messages



The screenshot shows a terminal window titled "ds@ds-VirtualBox: ~" with a dark background. The left sidebar contains icons for Activities, Terminal, Files, App Store, Help, and a terminal icon. The terminal output displays a series of attitude messages, each starting with a green "ds" prefix. Each message is a JSON object containing fields for time_boot_ms, roll, pitch, yaw, rollspeed, pitchspeed, and yawspeed. The messages are separated by newlines and some are preceded by a green "ds" prefix. The messages are as follows:

```
dsATTITUDE {time_boot_ms : 66375, roll : -0.029228951781988144, pitch : 0.23988442
[s12293625, yaw : -2.669882297515869, rollspeed : -0.006214757915586233, pitchspee
dsd : 0.002867966890335083, yawspeed : 0.01096766535192728]
dsATTITUDE {time_boot_ms : 66395, roll : -0.02903817594051361, pitch : 0.239900827
[s40783691, yaw : -2.6663219928741455, rollspeed : 0.0014906460419297218, pitchspe
dsed : 0.0006311186589300632, yawspeed : 0.009320607408881187}
[sATTITUDE {time_boot_ms : 66415, roll : -0.028971625491976738, pitch : 0.23995253
ds443717957, yaw : -2.6630842685699463, rollspeed : -0.0025250171311199665, pitchs
peed : 0.0004336046986281872, yawspeed : 0.006341955158859491}
ATTITUDE {time_boot_ms : 66432, roll : -0.028851941227912903, pitch : 0.23998275
39920807, yaw : -2.6607866287231445, rollspeed : 0.0018133415142074227, pitchspe
ed : -0.0003740927204489708, yawspeed : 0.004446226172149181}
ATTITUDE {time_boot_ms : 66452, roll : -0.028634045273065567, pitch : 0.24004840
850830078, yaw : -2.6577413082122803, rollspeed : 0.0024617179296910763, pitchsp
eed : -0.0006809770129621029, yawspeed : 0.004636307246983051}
ATTITUDE {time_boot_ms : 66472, roll : -0.028501328080892563, pitch : 0.24009604
75206375, yaw : -2.6549999713897705, rollspeed : -0.0027936664409935474, pitchsp
eed : 0.0006084633059799671, yawspeed : 0.0047083692625164986}
ATTITUDE {time_boot_ms : 66497, roll : -0.02838805690407753, pitch : 0.240179434
41867828, yaw : -2.6522746086120605, rollspeed : -0.005487538408488035, pitchspe
ed : -0.0005813487805426121, yawspeed : 0.010690251365303993}
```


Experiment 5: Ultrasonic Distance Readings



Experiment 6: Camera Direction Detection

